

Team Codebaes

Chrono-Architect

A Time Machine for Your Cloud Bill — AI-Driven AWS FinOps Agent

Team Leader: Rahul Gupta

Email: rahulmohan.gupta@mca.christuniversity.in

Hackathon: TrueFoundry Hackathon — Cloud Cost Janitor Track

1. The Problem

Cloud cost tools either dump a dashboard of unread alerts or let an automated script silently delete resources based on a guess. Neither approach earns trust: one produces alert fatigue, the other risks an outage. Teams need a system that investigates before it acts, shows its evidence, and only pulls the trigger once a human has weighed a clear, probability-scored choice.

2. Our Approach

Chrono-Architect is an evidence-driven FinOps agent that treats every cost-saving action — delete, right-size, or leave idle — as a decision to be proven, not guessed. A Manager Agent coordinates specialist agents to investigate a real cloud resource, simulate multiple possible futures in a sandbox, quantify the exact cost and blast radius of each option, and then pause at a cryptographically bound checkpoint for human approval before anything changes. No price, metric, or action in the system is hardcoded — everything is pulled live through the official AWS API MCP server.

The workflow for the first working slice is:

Investigate resource → gather evidence → build future scenarios → request human approval → recheck for drift → execute (stop / modify / start) → verify outcome.

3. Core Modules

- Manager Agent — orchestrates the specialist agents and the overall workflow
- Forensic Auditing — reconstructs how and why a resource reached its current state
- Real-Time Analysis — reads live metrics and configuration from AWS
- Future Simulation — models delete / right-size / idle outcomes via Monte Carlo-style runs in the TrueForge sandbox
- Cost Prediction — near-, mid-, and long-term cost projections using AWS pricing data
- Blast-Radius Analysis — maps what else in the environment is affected by an action
- Dependency Graph Analyser — checks what depends on a resource before it is touched
- Risk Scoring Engine — assigns a probability-weighted risk score to each option
- Human Negotiation / Approval — presents evidence and holds for an explicit, hash-bound sign-off
- Autonomous Rollback Module — reverts an executed action when approved to do so
- Continuous Monitoring Agent — watches resources and outcomes after execution
- Policy / Guardrail Engine — blocks unsafe actions (e.g. Spot, ASG-managed, production without a health check)
- MCP Integration — all AWS access goes through the AWS API MCP server; no direct hardcoded calls

4. Safety Model

Read and write access are separated into different worker identities. Only resources explicitly tagged as managed are eligible for action. Autoscaled, Spot, and instance-store resources are blocked outright, and production resources require a live health-check URL. Every approval is bound to the exact SHA-256 hash of the proposal it was shown and expires on a timer; the resource is re-read immediately before execution, and any drift invalidates the approval. All actions and evidence are written to an append-only, hash-chained audit log, and every calculation uses AWS-sourced data with exact decimal arithmetic rather than model-estimated numbers.

5. System Walkthrough

- Receive the user's request for a resource to evaluate
- Inspect the live cloud resource and its configuration
- Call AWS through MCP tools for metrics, pricing, and cost data
- Use an LLM (OpenAI, via TrueFoundry) for reasoning over the evidence
- Generate and run scenario analysis inside the TrueForge sandbox
- Compare delete / right-size / idle scenarios on cost and risk
- Stop and wait for explicit human approval
- Execute only the approved action
- Verify the result against fresh evidence

6. Operator Console (UI)

- Dashboard — overview of managed resources and recent activity
- Resources — inventory of cloud resources under agent management
- Analysis — evidence and forensic findings per resource
- Simulations — side-by-side future scenarios with predicted cost and risk
- Approvals — pending decisions awaiting human sign-off
- Monitoring — post-action tracking and rollback status
- Settings — model, MCP server, and skill configuration

7. Built on TrueFoundry

The agent, its OpenAI-backed reasoning model, and its MCP server / skill configuration are all set up through TrueFoundry. Each user prompt runs as a session, scenario comparisons run inside the TrueForge sandbox, and recurring checks (continuous monitoring) run as scheduled jobs. The same agent is also reachable through TrueFoundry's API for headless or automated use.

8. Current Scope and Honesty About Limitations

The working slice safely right-sizes one explicitly selected, standalone, EBS-backed, On-Demand EC2 instance end-to-end. List pricing is not a guarantee of realized billing savings — Savings Plans, Reserved Instances, private rates, and credits can change the real outcome, and native EC2 metrics do not include memory usage. Chrono-Architect is built to record these limitations and block on missing evidence rather than invent numbers to fill the gap.

9. Why It Matters

Chrono-Architect replaces a single risky guess with a forensic, evidence-backed, human-approved decision — turning cloud cost cleanup from a gamble into a predictable, explainable, and reversible process.